

Acanthosis Nigricans

In some cases, oral, esophageal, pharyngeal, laryngeal, conjunctival, and anogenital mucosal surfaces may be involved. In general, however, the back of the neck is the most consistently and severely affected area. The development of superimposed **acrochordons** in involved areas is well described. In particularly florid cases, involvement on the back of the hands over the knuckles and even on the palms can be seen. When the palms are involved, the rugated appearance of the palmar surface has been called **tripe palms** and is usually associated with acanthosis nigricans seen in the setting of malignancy.

In the majority of cases, the most important factor in diagnosing acanthosis nigricans is recognizing the usually associated **hyperinsulinemia**, which is a known risk factor for type 2 diabetes.

Histopathology

The histopathology of clinical lesions demonstrates **papillomatosis** and **hyperkeratosis** but minimal acanthosis. Hyperpigmentation of the basal layer has been variably demonstrated, and the brown color of the lesions is attributed to hyperkeratosis by most.

Treatment

Treatment of acanthosis nigricans is generally ineffective. Topical treatments with **calcipotriol**, **salicylic acid**, **urea**, and systemic or topical **retinoids** have all been used with anecdotal success. When identifiable, treatment of the underlying cause may be beneficial. Improvement or resolution does occur with **weight loss** in some obese patients. Medications that improve insulin sensitivity, such as **metformin**, have a theoretic benefit. Removal of an offending medication generally results in clearance of the skin.

In patients with acanthosis nigricans associated with malignancy, there is usually improvement following treatment of the underlying malignancy. The skin finding may present before or after the diagnosis of malignancy is made.

When associated with malignancy, a tumor of intraabdominal origin—usually **gastric**—is seen in the majority of cases. It has been repeatedly described that patients' skin improves with chemotherapy and relapses with recurrence.

Diabetic Thick Skin

Several specific syndromes are associated with localized thickening of the skin in diabetes. The common underlying pathogenesis involves biochemical alterations in dermal collagen and mucopolysaccharides. The clinical syndromes result from increased deposition and improper degradation of these constituents, likely related to the formation of **advanced glycation end products (AGEs)**.

Limited Joint Mobility and Scleroderma-Like Syndrome

Diabetic limited joint mobility (LJM), or **cheiroarthropathy**, presents as tightness and thickening of the skin and periarticular connective tissue of the fingers, resulting in a painless loss of joint mobility. Initial involvement typically begins at the distal interphalangeal joint of the fifth digit and progresses proximally to involve all fingers. Larger joints—such as those of the elbow, knee, and foot—may also be affected. However, the actual joint space remains uninvolved, so LJM is not a true arthropathy.

This disorder is characterized by the "**prayer sign**", which is the inability to fully approximate the palmar surfaces and interphalangeal joint spaces when the hands are pressed together with fingers extended.

In addition to joint contractures, the skin may appear **thickened, waxy, and smooth**, with apparent loss of adnexal structures, resembling the skin changes seen in scleroderma.

LJM occurs in **30% to 50% of adult patients with type 1 diabetes** and is also common in type 2 diabetes. It is associated with **longer duration of diabetes** and **poor glycemic control**.